From Random Hyperplanes to 71-Graph Evidence

A reproducible Max-Cut research program

SDP-inspired vector factors · spherical clustering · local search · G-set

Experiments 001–021 · 29 October 2026

Repository: heri-espino/maxkcut

Prepared for Professor Ángel E. Rodríguez Fernández

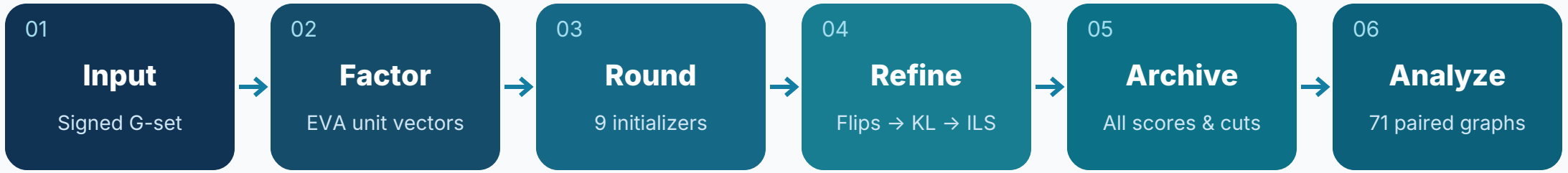
Final outcome at a glance

Scope	Verified result
Canonical G-set graphs	**71 / 71**
Initialization families	**9**
Matched restarts per graph and method	**50**
Fully archived Max-Cut searches	**31,950**
Missing factors / incomplete graphs	**0 / 0**
Graphs where best-of-nine beats best-of-control	**58 / 71**

Important: these are feasible **binary Max-Cut** results, **not** certified global optima or a general Max-k-Cut result for $k > 2$.

Source: Experiment021 report.

Research map: seven linked stages



One fixed vector factor per graph · 50 matched restarts · immutable saved partitions

One graph, one immutable vector factor, multiple comparable rounding strategies, exact re-scoring and archived outputs.

1. What problem did we solve?

From a discrete graph partition to a geometric representation.

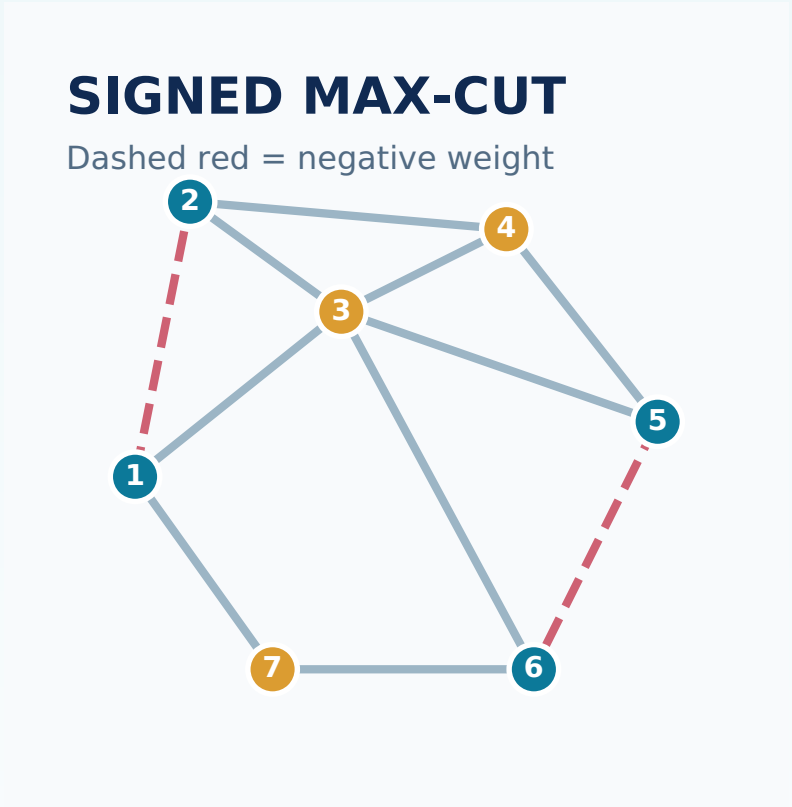
Formal objective: signed Max-Cut

For an undirected graph $G = (V, E, w)$, choose binary labels $c_i \in \{0, 1\}$:

$$\max_{c \in \{0,1\}^{|V|}} C(c), \qquad C(c) = \sum_{\{i,j\} \in E} w_{ij}\, \mathbf{1}[c_i \neq c_j].$$

- **Positive** w_{ij}: rewards placing endpoints on opposite sides.
- **Negative** w_{ij}: penalizes separating the endpoints.
- All reported cut scores use the **original signed weights**.

The clustering objective is **not** the cut objective: it proposes a partition that must subsequently be evaluated on the graph.



CONCEPTUAL SCHEMATIC

Classical SDP relaxation (the theoretical reference)

Let $L = D - W$ and $X = VV^\top$ be a Gram matrix. For general $k \geq 2$:

$$\max_{X \succeq 0,\, X_{ii}=1} \frac{k-1}{2k}\mathrm{Tr}(LX), \qquad X_{ij} \geq -\frac{1}{k-1} \quad (k > 2).$$

At $k = 2$, the pairwise lower bound is redundant. Factorization represents each vertex i by a unit vector v_i.

Crucial distinction: the large-scale campaign used **rank-restricted nonconvex EVA factors**; these are not proven solutions or certified bounds for the full convex SDP.

Source: repository `AGENTS.md`, `RESEARCH_HANDOFF.md` and Rodríguez Fernández (2021, Chapters 6–7).

Goemans–Williamson: random hyperplane rounding

Draw $r \sim N(0, I_p)$ and set

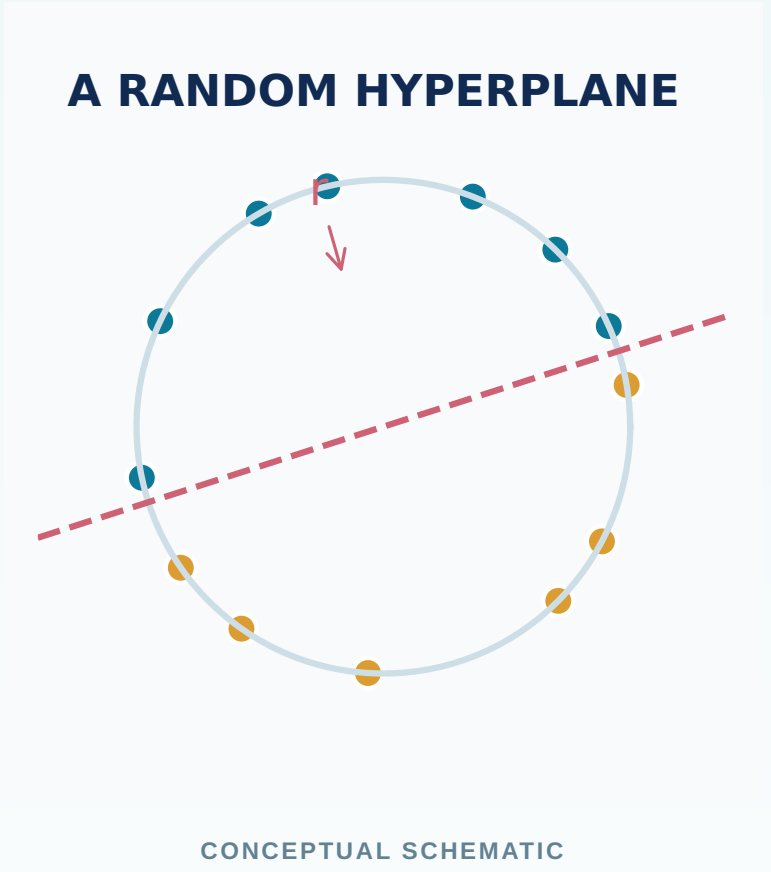
$$c_i = \mathbf{1}[v_i^\top r < 0].$$

The hyperplane $v^\top r = 0$ separates embedded vertices into two sides.

The starting question: why choose an arbitrary hyperplane if the vector embedding already contains structure from an optimization problem?

- Replace the initial two-way split with **spherical clustering**.
- Keep the original graph cut as the evaluation metric.
- Preserve the GW-equivalent starting cut in the original K-MeansNM control.

Source: Rodríguez-Fernández et al. (2020), Sections 1–3.



How spherical K-means enters Max-Cut rounding

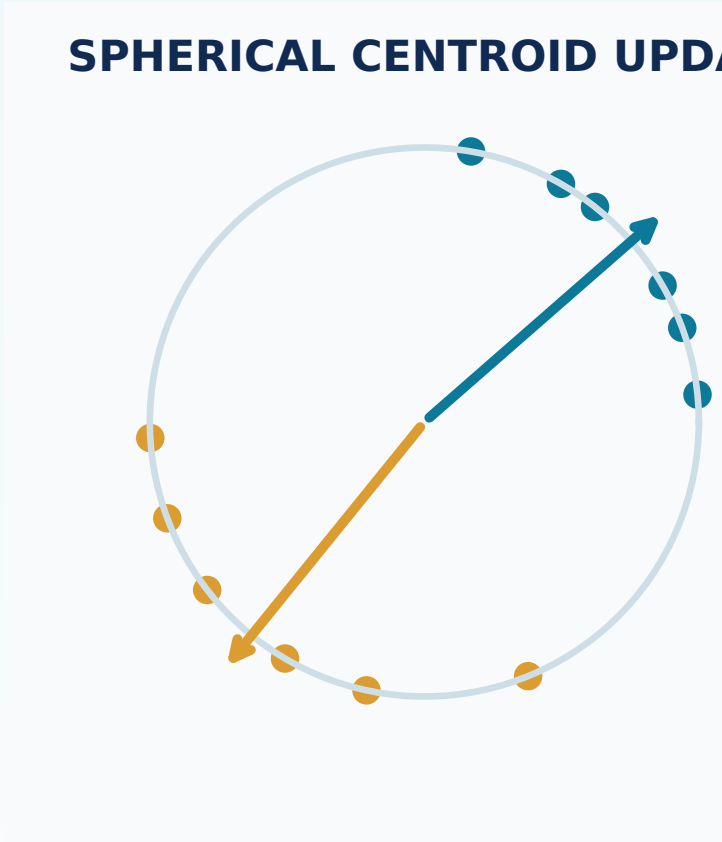
Original K-MeansNM initializes two unit centroids as u and $-u$. Its first assignment is the GW hyperplane partition.

Then alternate:

1. Assign each v_i to the centroid with maximum dot product.
2. Normalize each cluster's mean to update its centroid.
3. Keep the **best original-graph cut observed along the trajectory**.

Geometric distortion can decrease without the cut increasing at every step. This is why evaluating **intermediate** iterates matters.

Source: Rodríguez-Fernández et al. (2020), Eq. (4), Lemma 1 and Section 3.1.



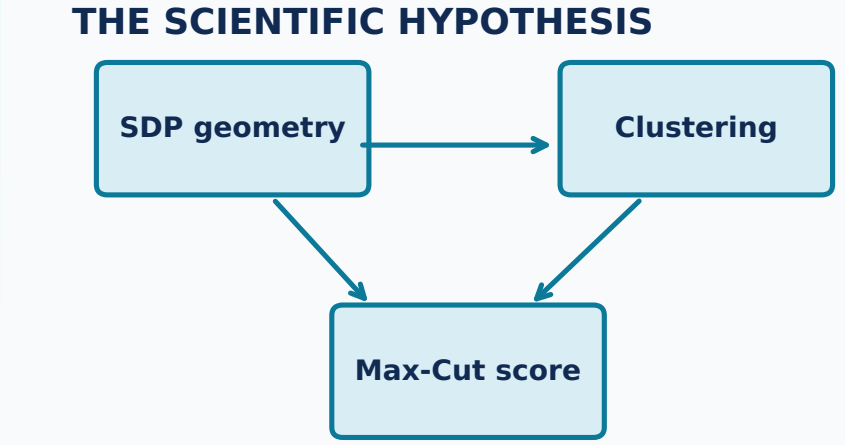
CONCEPTUAL SCHEMATIC

Three foundations from the literature

Work	What it contributed to our design
Rodríguez Fernández (2021), thesis	Max-Cut / Max-k-Cut, SDP geometry, fitness landscapes, clustering rounding
Rodríguez-Fernández et al. (2020)	K-MeansNM, K-Means2N and comparison with GW on G-set
Felzenszwalb, Klivans & Paul (2022)	Convex geometry of the k-way elliptope and iterative fixed-point rounding

The 2022 fixed-point method is **related prior art**, not a claimed component of our Experiment021 implementation.

Sources: project PDFs and research handoff.



CONCEPTUAL SCHEMATIC

2. How the research evolved

Prototypes, failed hypotheses, stronger refiners and a controlled comparison.

Experimental chronology: from groundwork to a stable pipeline

Experiments	Research direction	What we learned
001–003	I/O, full SDP tests, EVA vector factors and checkpoints	Separate exact graph scoring from numerical factor optimization
005–009	MST partitions, balanced cuts, forest coarsening	Geometry-only bipartitions can be too restrictive
010–013	Generalized KL, Tabu, cut-aware rounding, pipeline grid, ILS	Post-rounding search is essential; historic G63 best **26,917**
015–016	Direct hyperplane optimization; soft Bernoulli / entropy approaches	Alternative searches did not replace the strongest pipeline
017–018	Multi-group clustering, exact binary merging; nine initializers	Fix the downstream pipeline and test initialization systematically
019–021	Matched G63 experiment, 68-graph expansion, 71-graph completion	Produce a complete auditable statistical dataset

Numbering note: unlisted experiment IDs are not asserted to be completed.

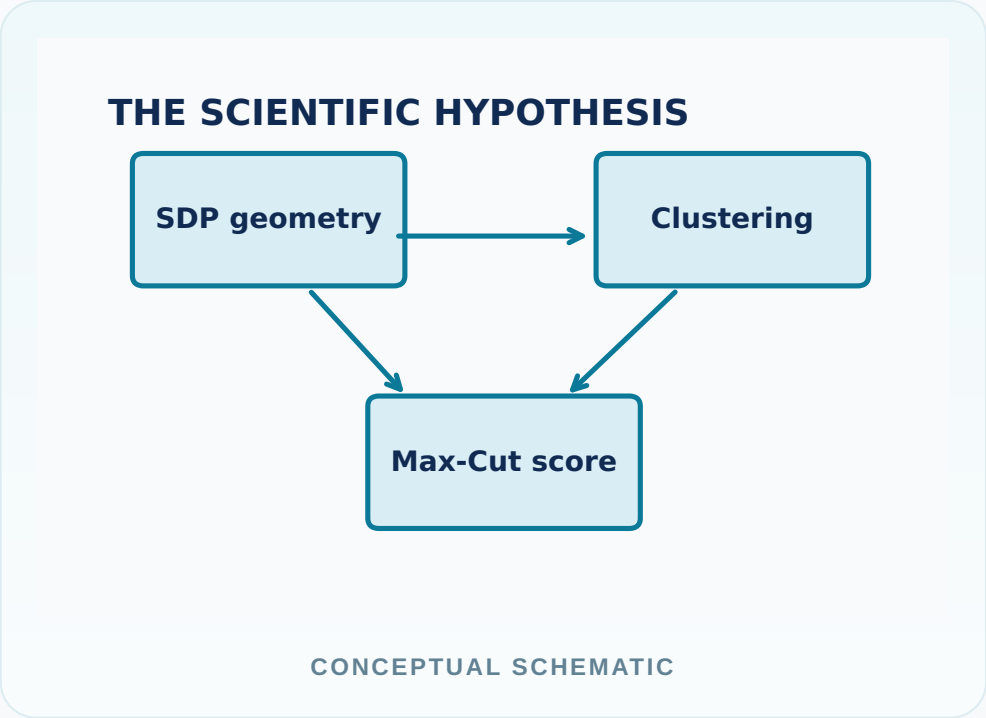
Source: `RESEARCH_HANDOFF.md` and the individual experiment notes.

Step 1 — Build a verifiable experimental substrate

- Keep canonical G-set text graphs, signed weighted edge lists and hashes.
- Implement numerical scoring separately from factor generation.
- Compare the small full-convex SDP pathway with scalable EVA approaches.
- Save reusable factors, iteration histories, seeds and binary partitions.
- Protect large jobs with explicit hardware/resource guards.

Why: we needed to distinguish a better *rounded cut* from a different embedding, input graph or solver approximation.

Repo: `src/maxkcut/{objective.py,sdp.py,eva.py,archive.py}`; Experiments001–003.



Step 2 — Obtain geometry with EVA

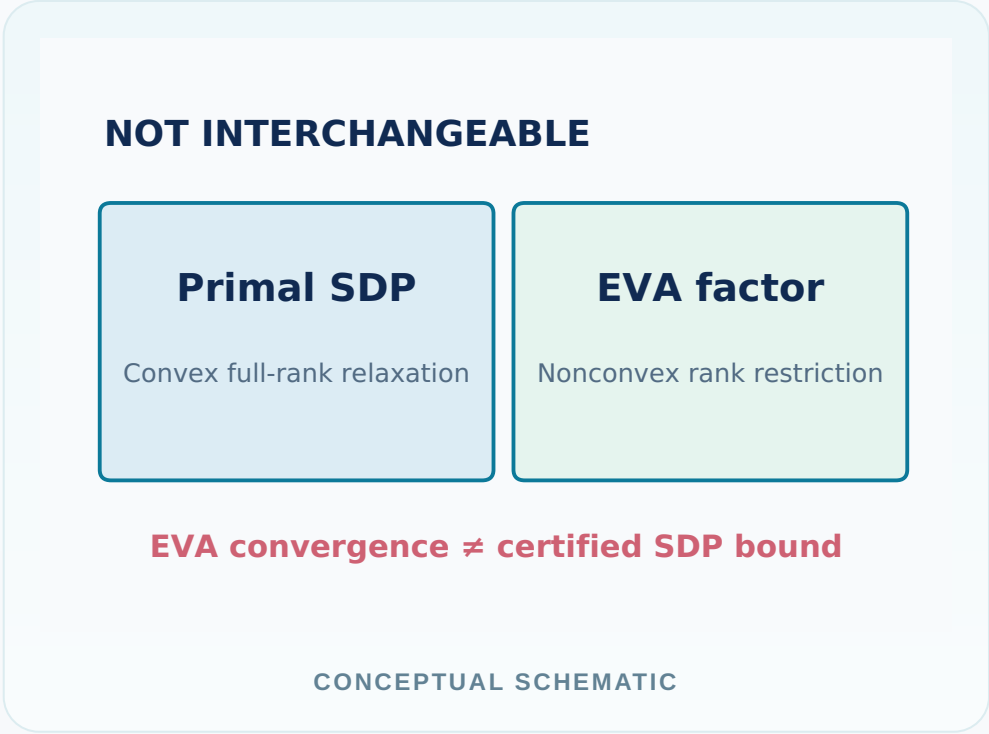
EVA optimizes a **rank-restricted vector relaxation** over row-unit embeddings $Y \in \mathbb{R}^{n \times p}$, with Y_i representing vertex i.

For the three final missing instances:

$$p = \lceil \sqrt{2(n+1)} \rceil = 119 \quad \text{for } n = 7000.$$

- Sequential Numba updates; one BLAS thread.
- Maximum 10,000 sweeps; angular tolerance **0.001 radians**.
- Seed **2026**; checkpoints every **25** sweeps.
- **No claim** of full-SDP convergence or global optimality.

Source: Experiment021 note and `.github/workflows/experiment021-complete-gset.yml`.



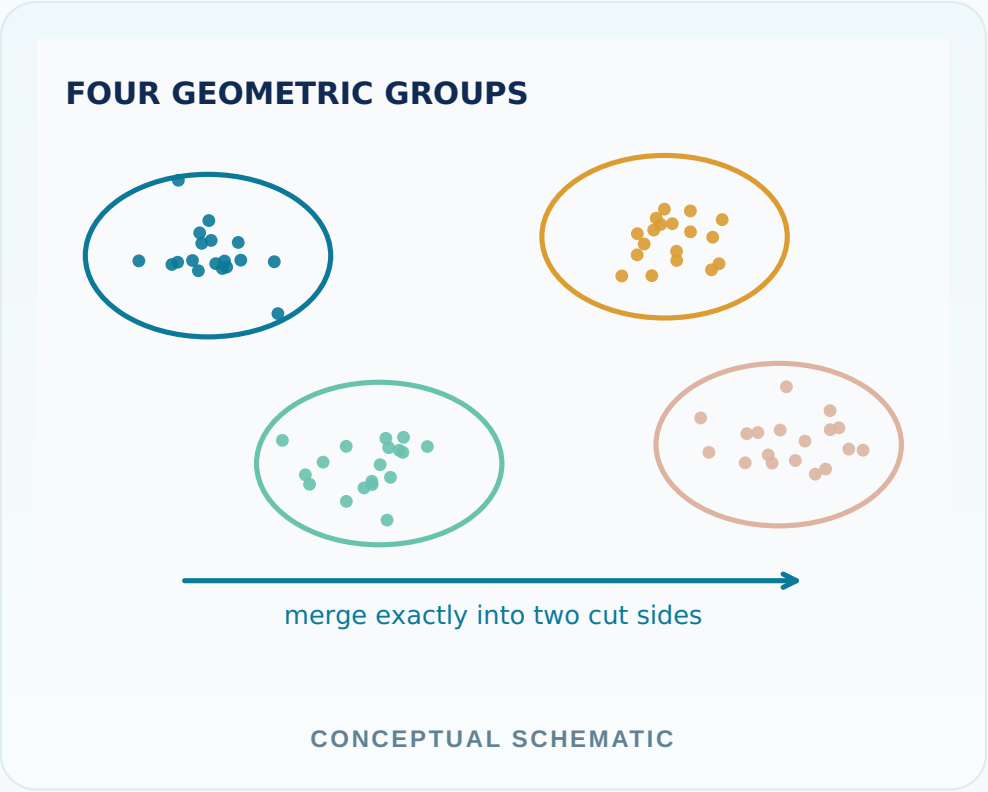
Step 3 — Test alternatives to naive rounding

MST approaches (005–009): full-dimensional Euclidean / other distances; tree-edge removals, multicomponent contractions, binary regrouping and vertex refinement.

Reason for moving on: rigid geometric clusters could miss good signed-graph cuts; G63's one-edge MST result was far below the later K-MeansNM + refinement baseline.

Critical invariant: when q MST edges are removed, the $q+1$ components are subsequently assigned back to **two** Max-Cut sides, not treated as $q+1$-Cut solutions.

Source: Experiment005 and Experiment009 notes.



Step 4 — Improve a cut with exact vertex gains

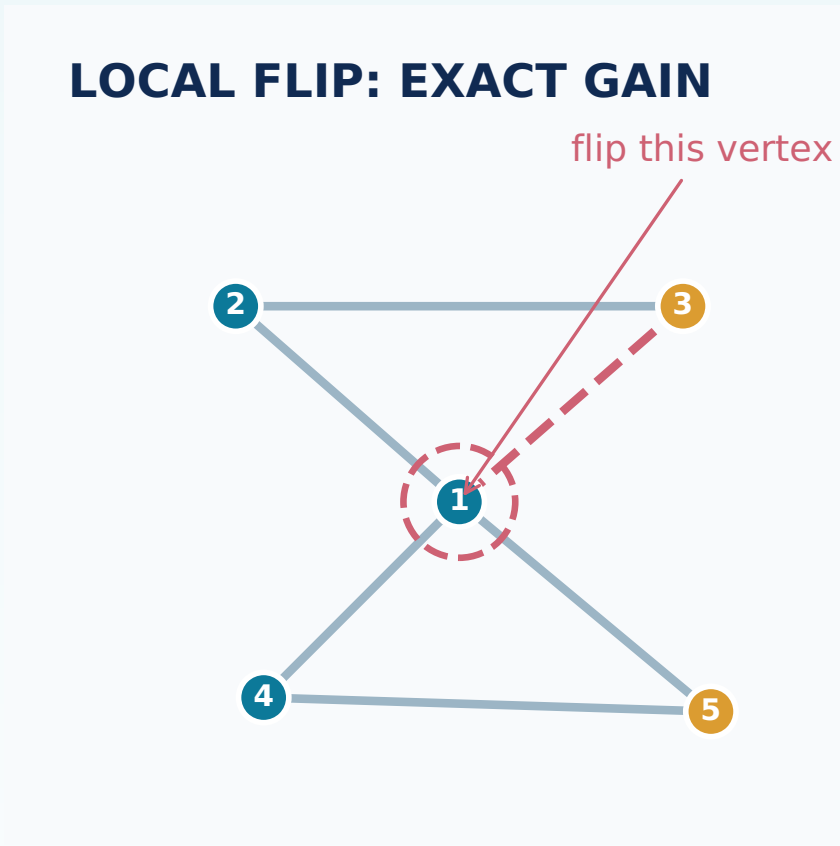
Represent side labels by spins $s_i \in \{-1, +1\}$:

$$C(s) = \frac{1}{2} \sum_{\{i,j\} \in E} w_{ij}(1 - s_i s_j), \qquad \Delta_i = s_i \sum_{j:\{i,j\} \in E} w_{ij} s_j.$$

Here Δ_i is the **exact change in cut** produced by flipping vertex i.

- **Flips:** accept positive-gain moves until locally stable or out of sweeps.
- Use the original signed sparse graph; update neighboring gains efficiently.
- Retain and independently rescore the resulting feasible partition.

Source: Experiment010 note, `src/maxkcut/` local-search implementation.



Step 5 — Escape the one-flip local optimum

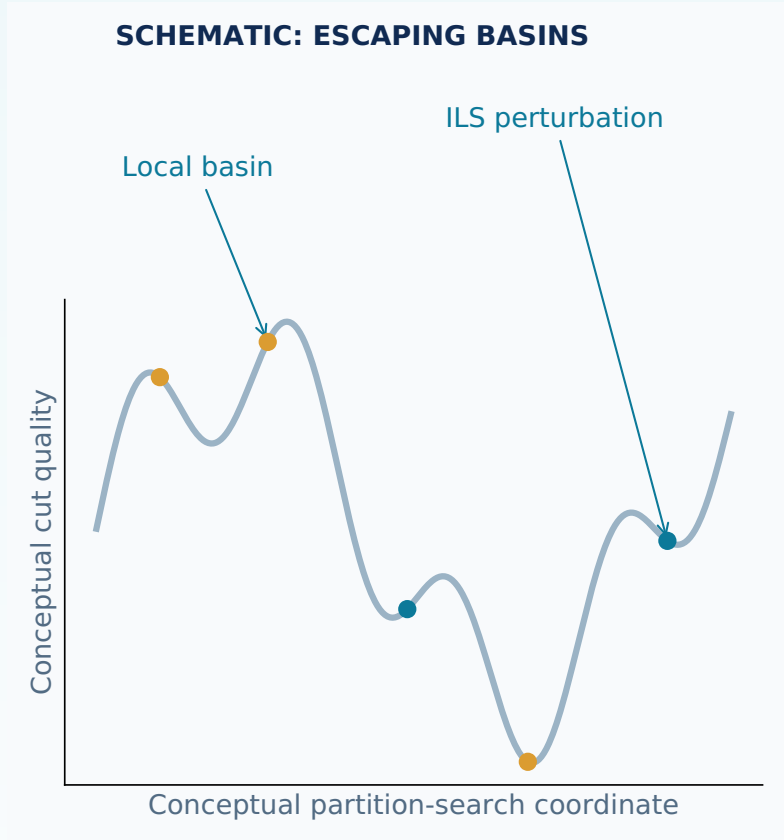
Generalized KL (not classical balanced pair-swapping):

1. In one pass, repeatedly choose the highest-gain unlocked vertex.
2. Permit temporarily negative gains.
3. Remember the best **strictly positive cumulative prefix**.
4. Accept that prefix; repeat up to the pass budget.

Iterated Local Search (ILS): perturb an elite partition, run inner KL, retain the best feasible cut visited.

This explains why the final score can be much stronger than the initial spherical clustering score.

Source: Experiments010 and 013; see Experiment019 pipeline.



CONCEPTUAL SCHEMATIC

Step 6 — Select the full refinement pipeline (Experiment013)

```
Frozen EVA vector factor
        ↓
Round with candidate method
        ↓
Positive-gain vertex flips
        ↓
Generalized signed KL
        ↓
KL-based iterated local search
        ↓
Final feasible signed Max-Cut + archived partition
```

The historically best verified **G63 feasible cut was 26,917**. The exploratory **27,047** target remained 130 higher and was **not** a proven optimum.

Source: `configs/experiment013-grid.json`, Experiment013 note.

Step 7 — Explore other models, but preserve the main baseline

Experiment	Hypothesis tested	Methodological distinction
015	A better rounding normal may be found directly	Exact great-semicircle event sweeps; stochastic smoothed-gradient ascent
016	Soft partition distributions may escape hard-rounding limitations	Entropy-regularized Bernoulli assignments, CEM/Gibbs proposals
017	More than two clusters may reveal useful latent structure	2/4/6/8/10 groups, **exact** merge back to binary cuts
018	Initialization of two centers may be the main lever	Nine systematic initializer families

For Experiment017, q geometric groups can be merged to two labels by enumerating $2^{q-1} - 1$ nontrivial complements; at $q = 10$, **511** mergers.

These experiments motivate **controlled initialization comparisons**, not a claim that every exploratory strategy improved the G63 record.

3. The frozen comparison protocol

Nine initializers, one embedding per graph, identical downstream budgets.

The nine initialization methods: part I

Name	How the initial centroids were selected
Original paired K-MeansNM	Random unit vector u and antipode $-u$
Independent centers	Two independently drawn random unit directions
Sample antipodal	An actual embedding row Y_i and $-Y_i$
Sample pair	Two different randomly chosen embedding rows
Farthest pair	One sampled row and a row maximally distant by cosine

All methods operate on the same graph-specific, **fixed EVA factor**.

Source: Experiment018 note and Experiment021 methodology.

The nine initialization methods: part II

Name	How the initial centroids were selected
Spherical K-means++	First sampled row; second selected by distance-squared weighting
Mean perturbation	Two perturbations around the normalized embedding mean
PCA antipodal	Leading centered principal direction + jitter, then antipode
GW screen-16	Evaluate 16 random candidate hyperplanes and choose the best

Budget warning: GW screen-16 uses **16 additional graph-cut evaluations** per restart, and PCA has preprocessing costs. Equal restarts does **not** mean equal compute.

Source: Experiment018 note.

Identical downstream refinement for every method

Stage	Frozen setting
Spherical Lloyd	$k = 2$, up to **100** iterations; keep best actual cut
Positive vertex flips	Up to **20** sweeps
Generalized KL	Up to **30** passes
KL-based ILS	**80** cycles; **8** inner KL passes
ILS perturbation sizes	**8, 16, 32, 64** vertices
ILS temperature	**12**
Restarts	**50** per graph and method

All scores are recorded at **Round → Flip → KL → Final**.

Source: Experiments019–021.

Reproducible RNG design: matched downstream seeds

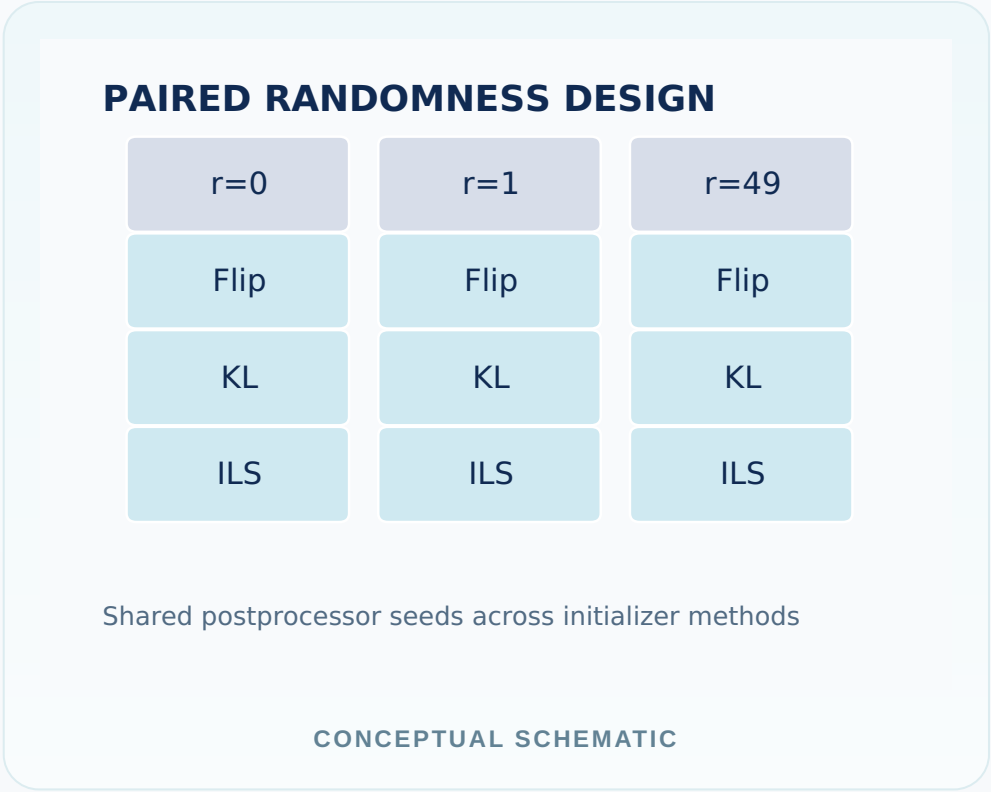
Graph index g, restart $r \in \{0, \ldots, 49\}$, seed base **2026**:

```
# Two-center rounding initialization
SeedSequence([2026, g, 2, r, 0])  # paired control
SeedSequence([2026, g, 2, r, 1])  # alternative methods

# Same postprocessing streams across methods, for matched r
flips_seed = [2026, g, r, 1313]
kl_seed    = [2026, g, r, 101]
ils_seed   = [2026, g, r, 1401]
```

Matched restart IDs control the **postprocessor randomness**, not the actual random initial centers drawn by different methods.

Source: Experiment019 and Experiment021 notes.

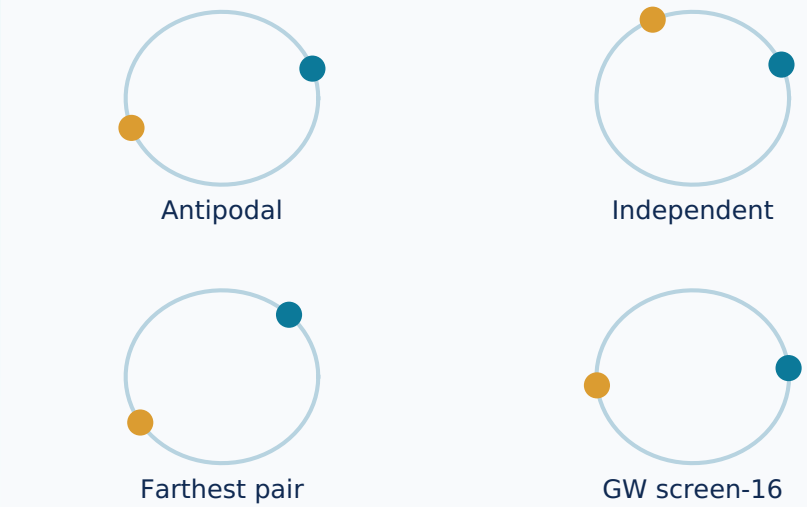


Experiment019 — Controlled pilot on G63

- Original G63 graph: **7,000 vertices**; fixed EVA factor with **119 columns**.
- **9 × 50 = 450** matched, full four-stage searches.
- Best new Experiment019 cut: **26,906** (farthest pair).
- Historical verified feasible record: **26,917** (Experiment013).
- Exploratory target: **27,047**, not a certified optimum.

Interpretation: the initializer pilot did not improve the historical G63 record, but gave a well-controlled design suitable for all G-set graphs.

Source: Experiment019 note; `RESEARCH_HANDOFF.md`.



4. From one graph to all 71

Windows self-hosted GitHub Actions, resumability and exact coverage checks.

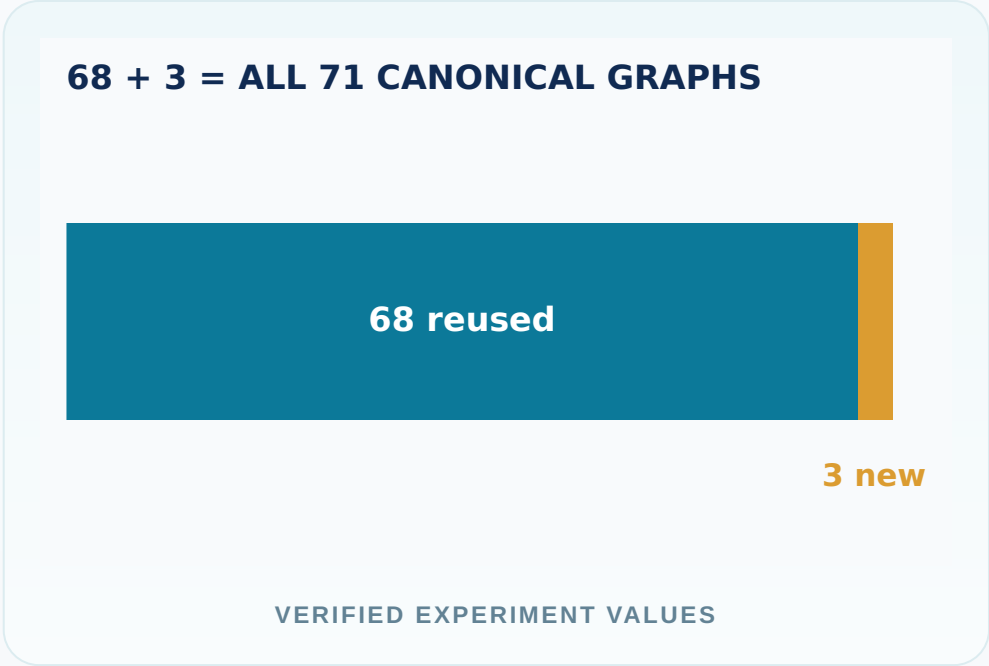
Experiment020 — Expand to the 68 graphs with saved factors

Goal: reuse already completed EVA embeddings; do not rerun the expensive factor solves.

- **68** eligible graphs with verified factors: **67 new** + reused G63.
- **68 × 9 × 50 = 30,600** archived full searches.
- **G60, G61, G62** initially excluded because matching completed factors were missing.
- **32** parallel Windows worker processes for rounding/ refinement.

A failed first workflow exposed a Windows worker race while writing JSON summaries. Recovery moved per-graph publication **after worker completion** and reused intact saved graph chunks.

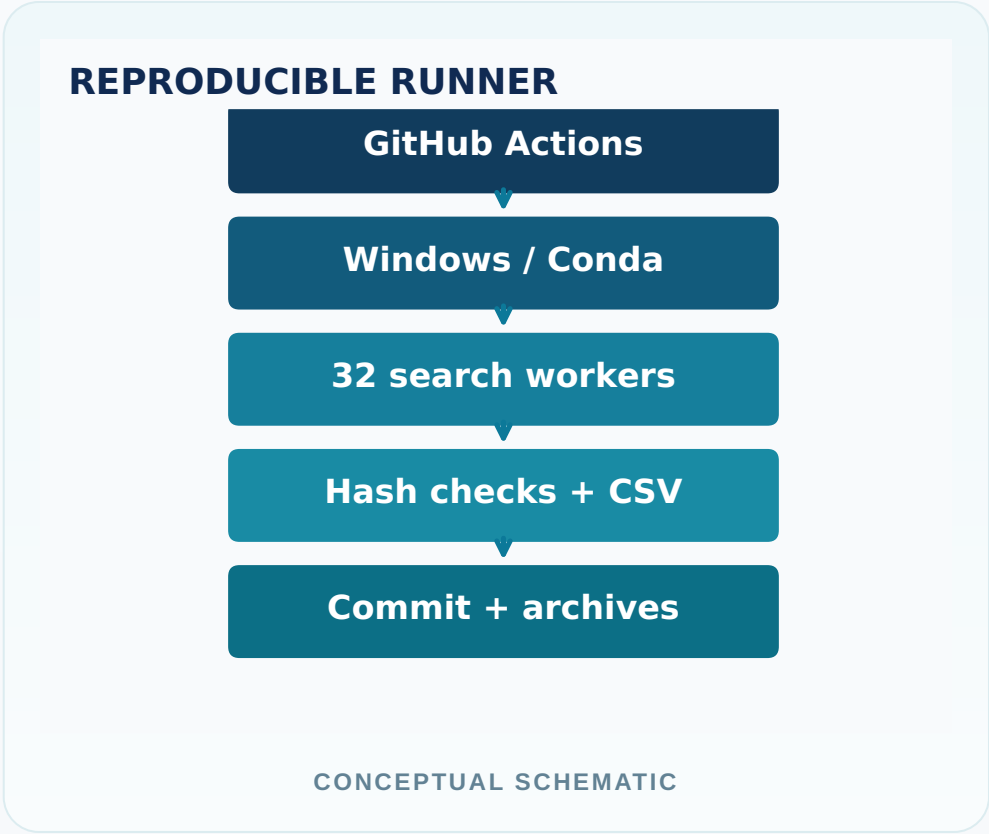
Sources: Experiment020 report, recovery run.



Why the self-hosted runner mattered

```
GitHub Actions dispatch / dedicated research branch
                  ↓
Windows 11 runner + maxkcut Conda Python
                  ↓
Factor/hash checks + fast regression tests
                  ↓
32 independent search workers; resumable chunks
                  ↓
Post-join aggregation and exact archive verification
                  ↓
Commit + push to experiment branch; upload artifact
```

Runner usage did not itself guarantee correct data: input hashes, process synchronization, restart completeness, seed identity and independent rescore checks were all required.



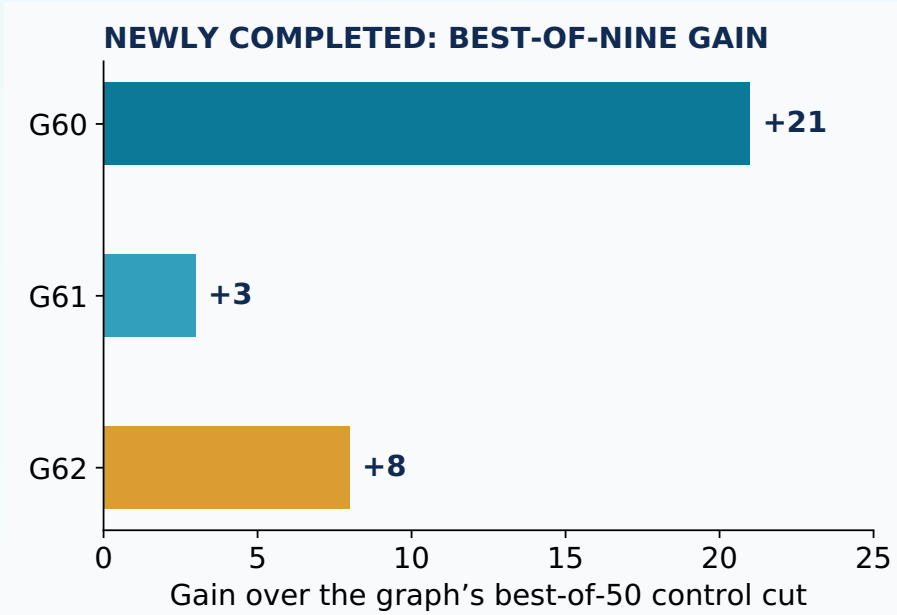
Experiment021 — Generate exactly the missing EVA factors

Graph	Vertices	Edges	EVA sweeps	Recorded state
G60	7,000	17,148	239	converged
G61	7,000	17,148	235	converged
G62	7,000	14,000	571	converged

Only these three new nonconvex factors were generated. Original Experiment019/020 factor files and search results remained unchanged.

Then: 3 × 9 × 50 = **1,350** new Round→Flip→KL→ILS searches.

Source: Experiment021 note.



VERIFIED EXPERIMENT VALUES

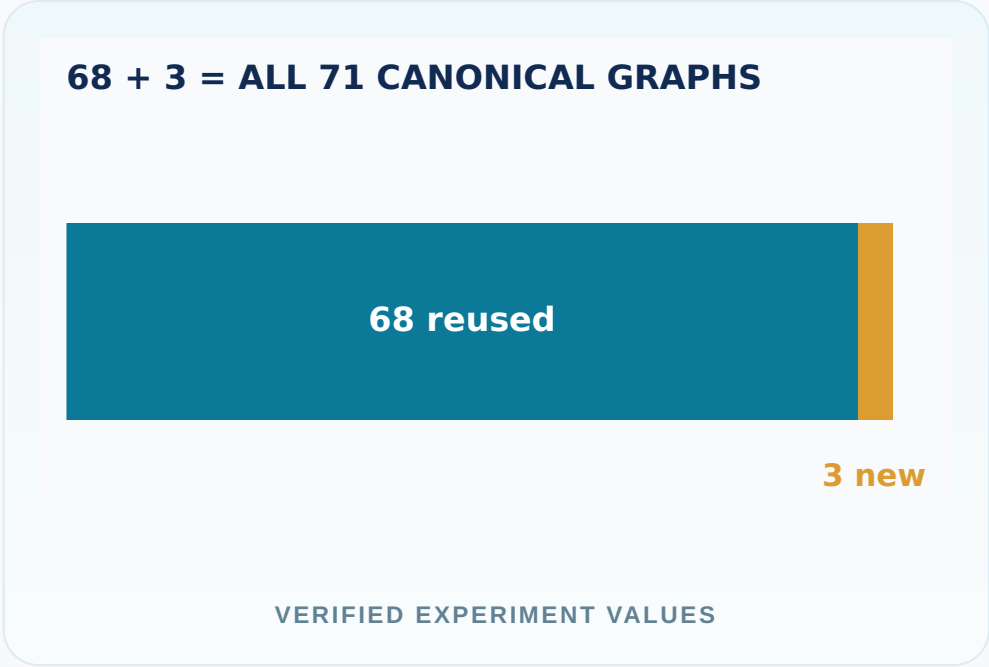
Final completeness proof is a machine-checked ledger

$$\underbrace{68 \times 9 \times 50}_{30{,}600\ \text{reused}} + \underbrace{3 \times 9 \times 50}_{1{,}350\ \text{new}} = \boxed{31{,}950}.$$

The consolidated report requires:

- Exactly **71** canonical graph identities.
- Exactly **nine** methods and **50** restart IDs per graph.
- **Zero** missing factors and **zero** incomplete instances.
- Original graph and factor **SHA-256** identities intact.
- Complete saved scores, partition files and per-restart traces.

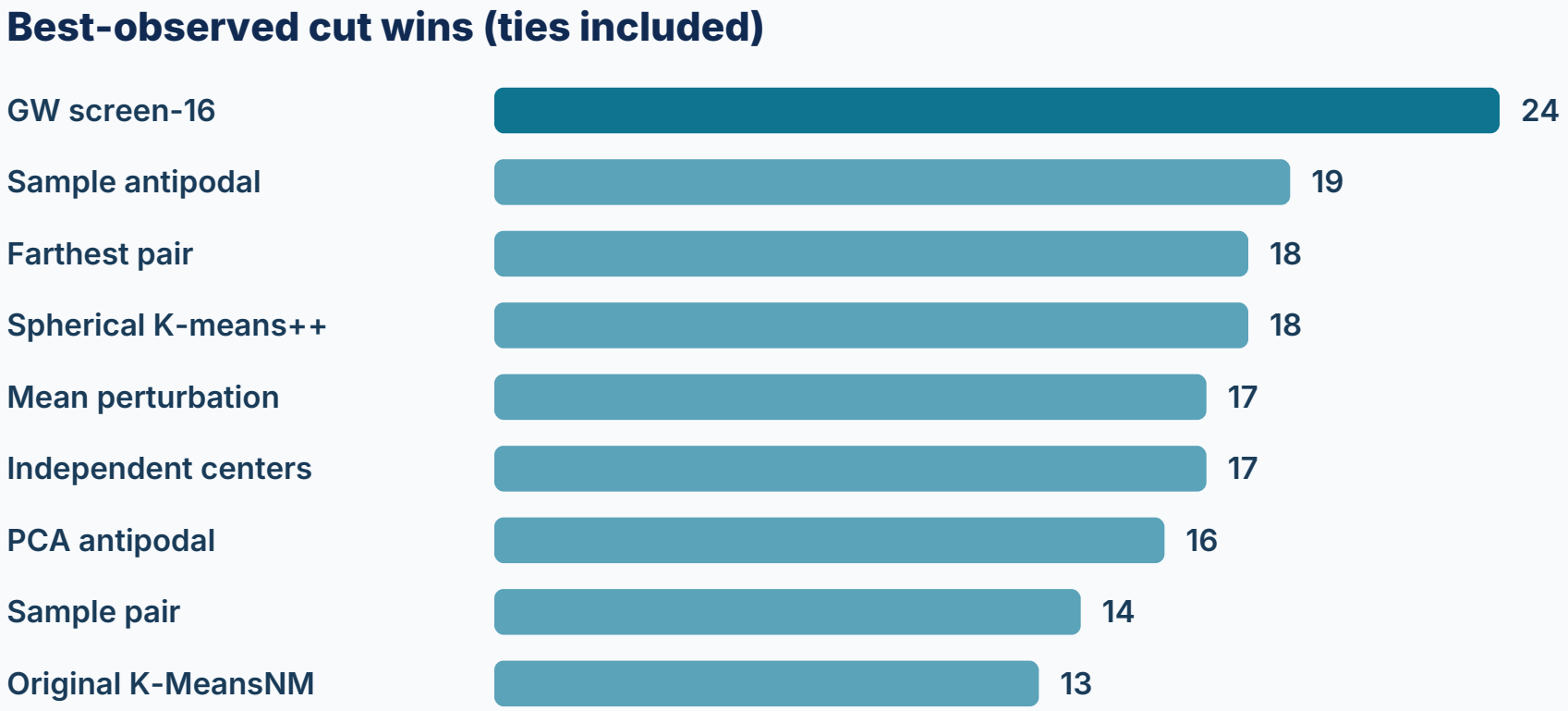
Passed: GitHub Actions #40831249367.



5. Results and interpretation

Distinguishing best individual cuts from average performance.

Ranking A — Who reached the best cut on most graphs?



Best-cut ties count for all tied methods. Values are graph counts out of 71; they need not sum to 71.

Source: `statistics/per_instance_methods.csv`, recomputed from `best_final`.

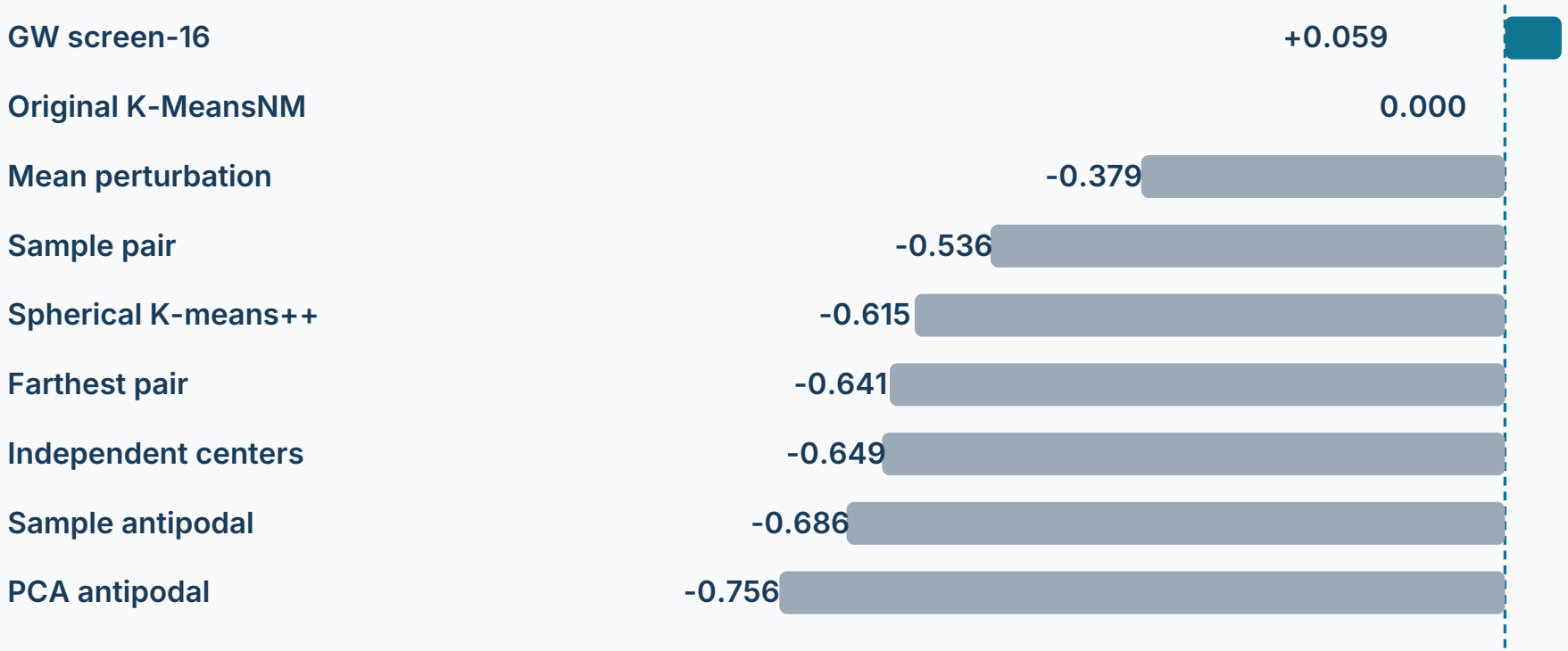
Ranking B — Share credit when several methods tie

Rank	Initialization	Graphs best (incl. ties)	Sole best	Fractional wins
1	**GW screen-16**	**24**	**7**	**12.19**
2	Farthest pair	18	**7**	9.46
3	Sample antipodal	19	5	8.75
4	Mean perturbation	17	6	8.50
5	Independent centers	17	4	7.21
6	Sample pair	14	4	6.82
7	Spherical K-means++	18	2	6.25
8	Original K-MeansNM	13	4	5.92
9	PCA antipodal	16	3	5.91

Fractional win = $1/t_g$ for each best-scoring method on graph g, where t_g is the number tied for best. The fractional wins sum to **71**.

Ranking C — Average final-cut difference versus control

71-graph mean final-cut difference vs original K-MeansNM



Unweighted average of each graph's 50-restart paired mean differences · original signed cut units

Method: calculate each graph's 50-restart mean difference relative to original K-MeansNM; then average the **71 graph-level differences** equally. Positive is better.

Three newly completed instances: best observed cuts

Graph	Best initializer	Best of nine	Control best	Gain
G60	Sample antipodal	**14,124**	14,103	**+21**
G61	Independent centers	**5,721**	5,718	**+3**
G62	Mean perturbation	**4,816**	4,808	**+8**

All values are **feasible signed Max-Cut scores**, not certificates of optimality.

Source: Experiment021 finalized per-graph records.

Important graph-specific landmarks

G63

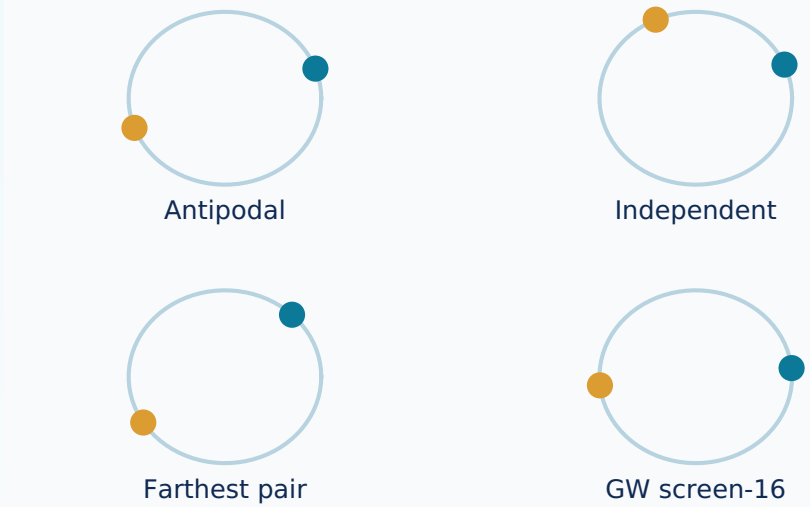
- Historical best observed feasible cut: **26,917** (Experiment013).
- Experiment019's nine-initializer best: **26,906**.
- Distance from historical record to exploratory **27,047** target: **130**.

G11

- A search attained **564** on the canonical instance, matching the repo's cited established optimum for that weighted graph.

Do not transfer historical targets, cut scales or certified-optimum labels between different graph identities.

Sources: `RESEARCH_HANDOFF.md`, Experiment019 note, `docs/gset-reference-values.md`.



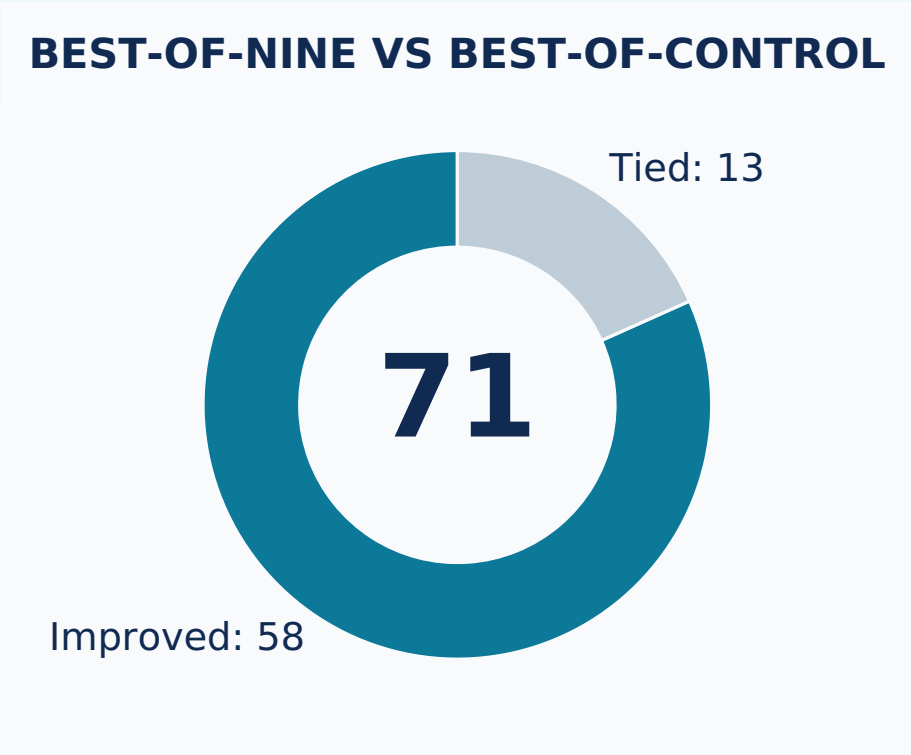
Why 58/71 is not an equal-budget victory

The best of nine methods beat the best of 50 control restarts on **58 graphs** and tied on **13**.

But it had up to **450 searches per graph**, versus **50** for the original K-MeansNM alone.

A portfolio containing the control can never score worse in best-of-search, by construction. Its 58 victories are empirical evidence of *diversity and opportunity*, not proof that an individual method is more effective per unit computation.

Also, GW screen-16 considers 16 additional initial hyperplanes per restart.



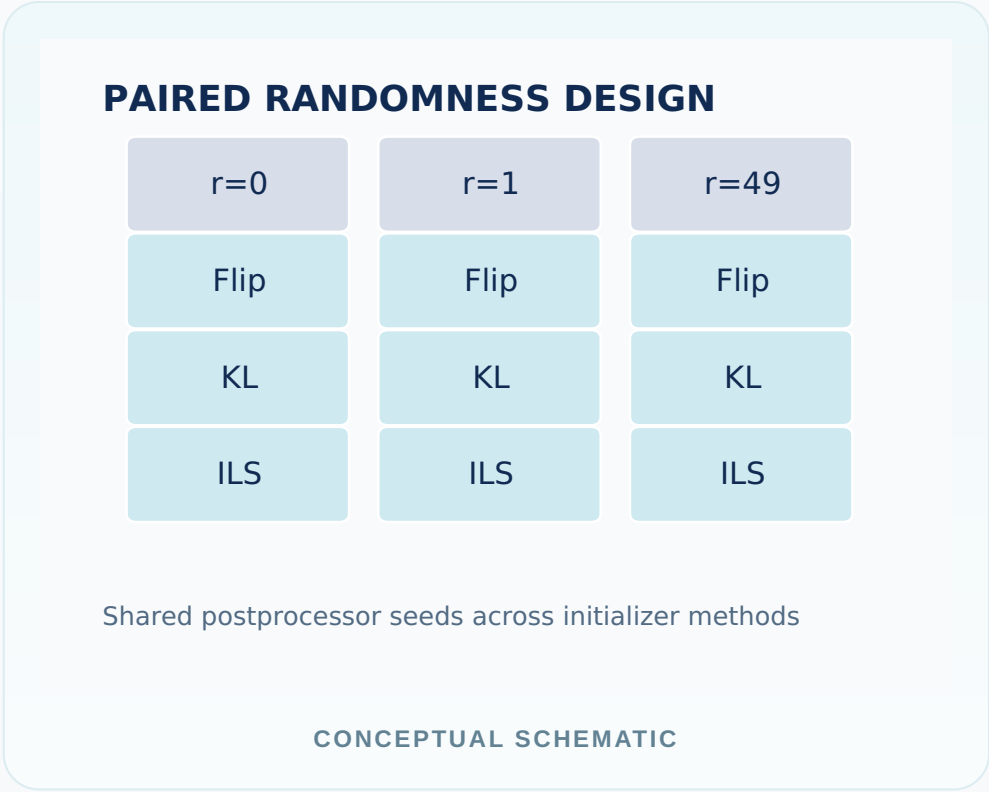
What the statistics do—and do not—show

Paired comparison at graph g:

$$\Delta_{g,m} = \frac{1}{50} \sum_{r=0}^{49} \left[C^{\text{final}}_{g,m,r} - C^{\text{final}}_{g,\text{control,r}} \right].$$

- Report the mean of the **71** graph-level $\Delta_{g,m}$ values.
- Within graph, **4,000-replicate paired bootstrap** intervals examine restart variability.
- Do **not** treat 31,950 correlated restarts as independent graphs.
- Do **not** equate a best-of-50 score with expected performance.
- No universal approximation ratio follows from these cut scores.

Source: Experiment021 note; `paired_vs_original_kmeansnm.csv`.



6. Reproduction and next steps

What to verify, where the evidence lives, and what remains open.

Repo-first evidence trail

Path in `maxkcut`	Role
`data/G*.txt`	Canonical 71 weighted graph inputs
`src/maxkcut/`	Factor, rounding, refiners, aggregation code
`results/gset/experiment019-.../`	Historical G63 full-run records
`results/gset/experiment020-.../`	67 additional graphs + analysis
`results/gset/experiment021-.../`	G60–G62 factors, cases, archives, full 71-graph statistics
`notes/experiments/`	Why/how/what happened/what next
`AGENTS.md`, `RESEARCH_HANDOFF.md`	Reproducible AI-agent handoff

The **GitHub branch**, not the runner's ephemeral scratch space, holds the research record.

Exact verification without repeating the expensive searches

From the populated Experiment021 branch, in Windows PowerShell:

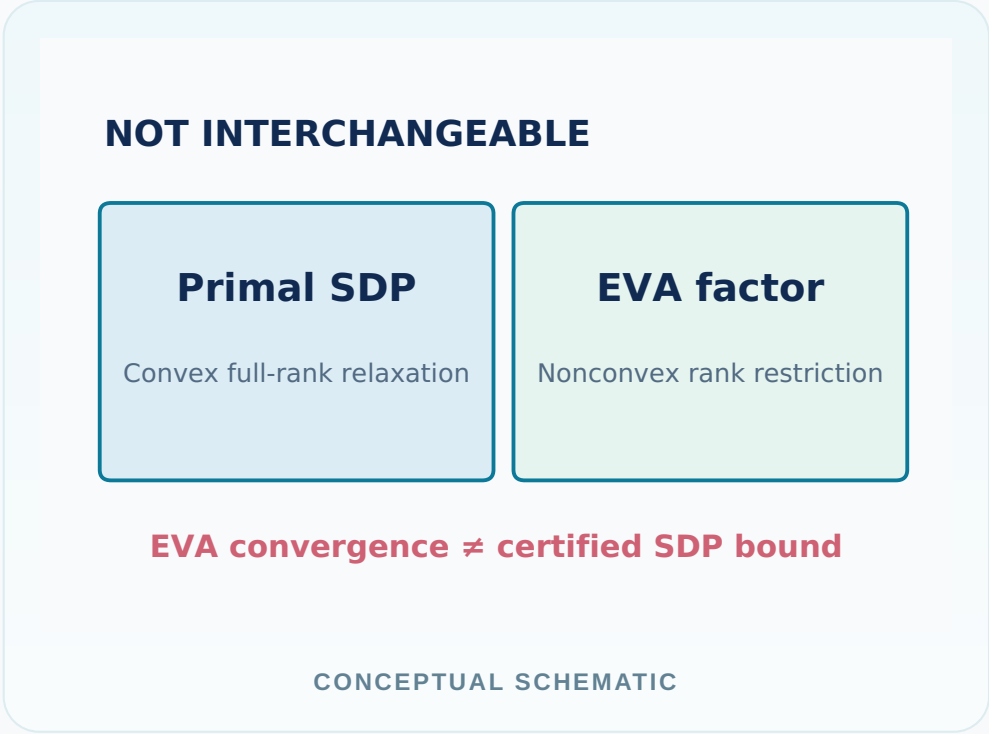
```
python -m pytest tests/test_eva_full_gset_completion.py tests/test_eva_kmeans_init_all.py tests/
test_eva_kmeans_init_full.py -q
python -m maxkcut.eva_full_gset_completion_cli inventory
python -m maxkcut.eva_full_gset_completion_cli aggregate
Get-Content "results/gset/experiment021-complete-71-gset/statistics/report.json"
```

The aggregator raises an error if a factor, graph, method or restart is missing, or if the total is different from **31,950**.

Do not rerun completed EVA/rounding just to regenerate the report.

Experimental limitations

1. EVA is **rank-restricted and nonconvex**; factor convergence is not an SDP certificate.
2. The study concerns **Max-Cut** $k = 2$, not a full sweep over $k > 2$.
3. Raw cut magnitudes differ by graph; report within-graph comparisons.
4. GW screen-16 and PCA do not have equal initialization overhead.
5. Best-of-nine portfolios have larger budgets and exploit selection.
6. Bootstrap intervals across starts do not establish graph-universal inference.
7. G63's **27,047** remains an exploratory target rather than a verified optimum.



Experiment022 — The next testable question

Hypothesis: a *diversified* initializer portfolio can exceed repeated K-MeansNM when total computational costs are fixed.

Pre-register before the next run:

- Same graph-specific EVA factor, signed weights and refinement protocol.
- Equal **wall-time and initial-cut evaluation budgets** per graph.
- Compare control-only, GW screen-16, and mixed-method portfolios.
- Evaluate graph-normalized improvement, best-of-budget and stability.
- Use paired graph-level statistics, multiple-comparison corrections and held-out design choices.

Final synthesis

Structure-aware initialization diversified the solutions we found; the original antipodal K-MeansNM remains a strong baseline.

We now have a verified, reproducible **71-graph / 31,950-search** dataset to test algorithmic efficiency—not just record cuts.

Primary sources and research record

Literature

- Rodríguez Fernández, A. E. (2021). *On the Study of Fitness Landscapes and the Max-Cut Problem*. Dissertation, Chapters 6–7.
- Rodríguez-Fernández, A. E., et al. (2020). *Clustering Improves the Goemans–Williamson Approximation for the Max-Cut Problem*. **Computation 8**, 75. DOI: 10.3390/computation8030075.
- Felzenszwalb, P., Klivans, C., & Paul, A. (2022). *Clustering with Semidefinite Programming and Fixed Point Iteration*. **JMLR 23**, article 21-0402.

Repository records

- Research handoff
- Experiment019 · Experiment020 · Experiment021
- Machine-readable statistics

Status: PR #50 and PR #51 were unmerged when this deck was prepared.